



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Malaysia-Japan
International
Institute of Technology
(MJIT)

WEB TECHNOLOGY PROJECT ASSIGNMENT: INCIDENT REPORTING SYSTEM (SECJ3483-15 WEB TECHNOLOGY)

LECTURER: DR. MUHAMMAD AIMAN BIN MOHD ZUKI

NAME: KHAIRUL FAZLI BIN MOHD YUSOF

MATRIC: A23MJ5091

SECTION: 15

NOTE: NO RPA IS IMPLEMENTED IN THIS PROJECT

1 INTRODUCTION

1.1 Background

This project is a collaboration with the real industry, DHL Global Finance Services, as DAC 3.0 Challenge. The project tackles real-world problems faced by DHL's Customer Support teams. receive a high volume of incident reports every day; late deliveries, address issues, damaged parcels, system errors, and customer complaints. These reports come through multiple inconsistent channels, such as

DHL Customer Support and Logistics Operations teams receive many incident reports every day involving late deliveries, damaged parcels, address issues, system errors, and customer complaints. These reports come from different communication channels such as emails, WhatsApp or Teams messages, phone call notes, screenshots, and handwritten instructions.

Because the information is unstructured and scattered across multiple platforms, incident management becomes difficult and inefficient. Therefore, this project proposes a web-based Incident Reporting System that centralizes incident reporting, tracking, and management into a single platform for easier monitoring and faster response.

1.2 Problem Statement

Currently, DHL manages incident reports manually through different communication channels and documents. This causes problems such as inconsistent report formats, duplicated incidents, difficulty tracking incident progress, and slow assignment to the correct department.

In addition, management teams face difficulties generating reports and monitoring incident status because the information is not centralized. These issues reduce operational efficiency and may affect the quality of customer support services.

1.3 Proposed Solution

To solve these problems, this project proposes a web-based Incident Reporting System for DHL Scenario 2. The system allows authorized users to upload, manage, search, and track incident reports through a centralized platform.

The system includes features such as secured login, incident status tracking, searching and filtering, and structured incident storage using a JSON database. By implementing this solution, DHL can improve incident management efficiency, reduce manual work, and provide faster and more organized customer support operations.

2 SYSTEM ARCHITECTURE

2.1 Front-End

Incident Reporting System is developed by HTML to structure the webpages and webpage content, decorated by CSS stylings for visual improvements and JavaScript that enables responsive interactive elements (like buttons) between the user and the website without needing to reload page.

Incident Reporting System includes

- **Login Page:** Authenticates user by username and password before granting access to website
- **Dashboard Page:** Main webpage that displays the summarised status of incidents

- **Upload Page:** Allows user to upload text files into the system that automatically creates new incident report
- **Incident Logging Page:** Displays all incident reports in structured list format where users can search for specific ones, filter by attributes and sort the list in ascending or descending order
- **Incident Information Page:** Displays full information of the selected incident report, also has *editing mode* for user to edit some data of the incident report and allows user to delete selected incident report

2.2 Back-End APIs & Database

Incident Reporting System implements noSQL-style JSON database that is hosted by Node.js to store the core data in the system; **users** and **incidents**, as object-based structure than common relational database.

Application Programming Interface (APIs) files constructed by combination of JavaScript and Node.js act as back-end components that handle system processing, data management and communication between the database and the front-end.

The following API files are

- **loginAPI.js:** Handles user authentication and login validation using `localStorage.setItem()`
- **mainAPI.js:** Retrieves the statuses of all incidents, displays as simple dashboard and also has navigations between modules
- **uploadAPI.js:** Processes incident upload files as new **incidents**-object in the JSON database
- **logsAPI.js:** Retrieves and manages all incident and displays in the logging page
- **infoAPI.js:** Retrieves detailed information for selected incident via `localStorage.getItem()` and processes the editing or deletion into the JSON database

3 System Structure

3.1 File Structure

File structure shows the project directory.

Bold represents folder, unbold represents file.

- **api**: Folder that stores all backend logic in .js
- **css**: Folder that stores all CSS styling files to the front-end
- **data**: Folder that has db.json which is the main system database
- **html**: Folder that stores all website pages in .html
- **node_modules**: Folder that stores all dependencies for Node.js database server implementation
- **resource**: Folder that stores all image resources for the website CSS styling
- package-lock.json: Dependency for .json database
- package.json: Also dependency for .json database

Root Folder	Folder Layer 1	Folder Layer 2
project-dhl2	api	infoAPI.js
		loginAPI.js
		logsAPI.js
		mainAPI.js
		uploadAPI.js
	css	info.css
		login.css
		logs.css
		main.css
		upload.css
	data	db.json
	html	info.html
		login.html
		logs.html
		main.html
		upload.html
	node_modules	...
	resource	filtercone.png
		filter.png
		sortASCEND.png
		SortDESCEND.png
	package-lock.json	-
	package.json	-

3.2 Database Structure

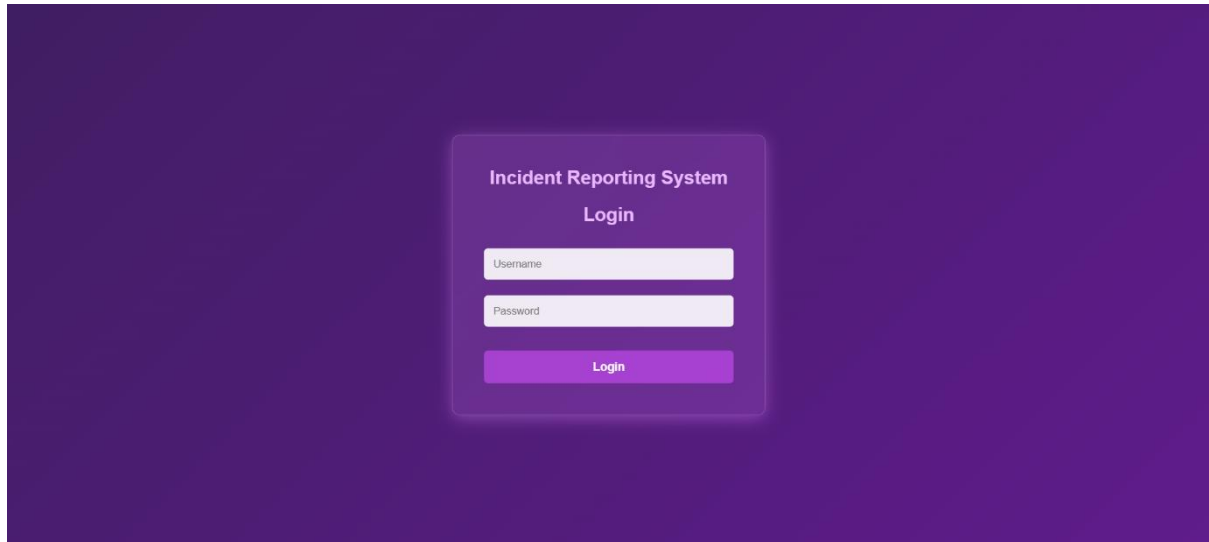
Object data is stored in noSQL-type database in *db.json* file.

Object	Attribute	Description	Type
users	id	Unique id of user	String
	username	Unique username of user	String
	password	Password of user	String
incidents	id	Unique id of incident	String
	creator	Username that uploaded the incident	String
	date	Date the incident uploaded	String
	status	Status of the incident (Draft -> Reviewed -> Published)	String
	description	Description of the incident (entire .txt file uploaded as description)	String
	tag	Tag of the incident	String
	department	Department to manage the incident	String
	channel	Source of the incident	String

4 WEBSITE INTERFACES

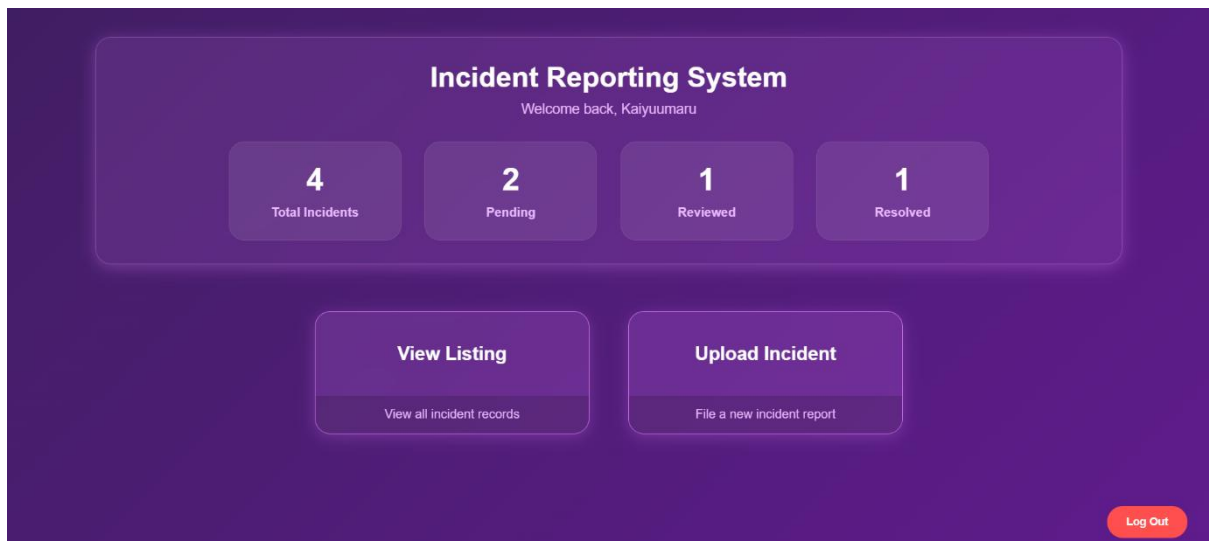
4.1 Login Page

Authentication with username and password before accessing website.



4.2 Dashboard Page

Main dashboard that displays the status of the uploaded incident reports in the system.



4.3 Upload File Page

Upload any .txt file from local device into the system to create new incident report.

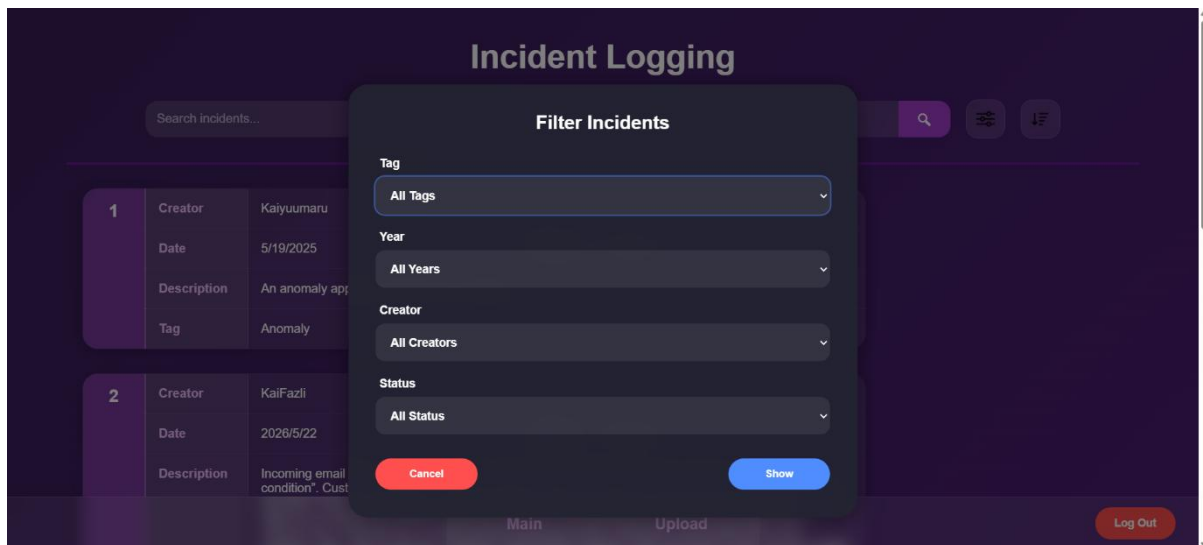
4.4 Incident Logging Page

View all existing incident reports in the system.

Has helper functions; search by any matching string, filter by **Tags**, **Year**, **Creator**, **Status** and lastly sort the list in ascending or descending ID order.

ID	Creator	Date	Status	Description
1	Kaiyuumanu	5/19/2025	Published	An anomaly appeared and stole my package.
2	KaiFazli	2026/5/22	Reviewed	Incoming email from customer on 22 March with subject: "Damaged parcel received – unacceptable condition". Customer attaches 4 images showing a crushed DHL box, torn side panels, and bubble wrap

Filter option

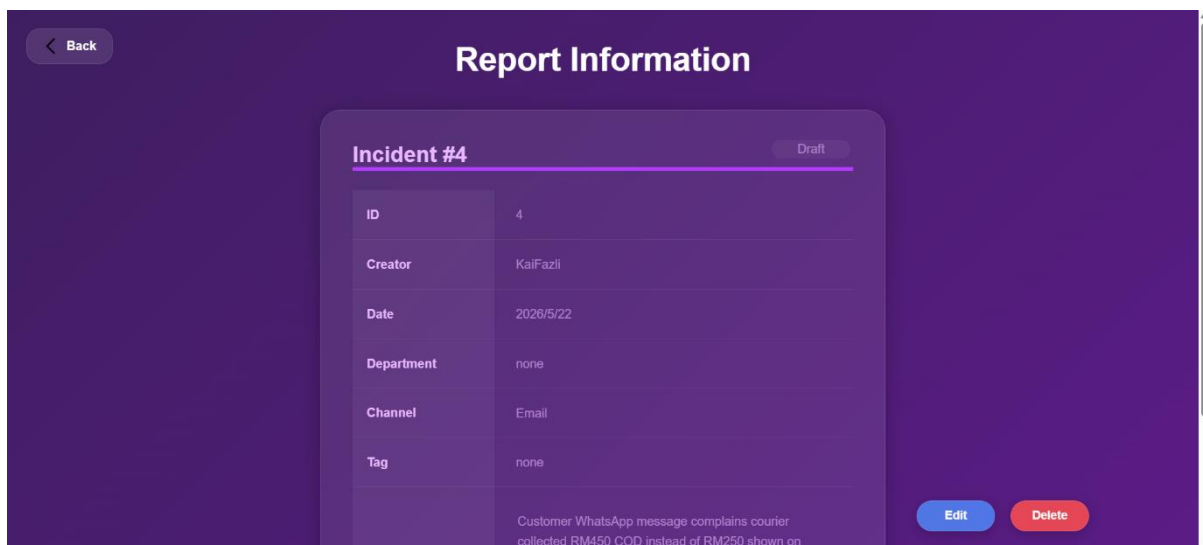


4.5 Incident Information Page

View the detailed version of the incident report (department and channel included).

"Edit" to enter *editing mode* to update the report.

"Delete" to remove the report from the system.



Editing mode: in editing mode, **Date, Department, Channel, Tag, Status** can be rewritten.

< Back

Report Information

Incident #4

Draft

ID	4
Creator	Kaifazli
Date	2026/5/22
Department	none
Channel	
Tag	none

Customer WhatsApp message complains courier collected RM450 COD instead of RM250 shown on

Cancel

Confirm

5 FUTURE PLANS OF RPA

5.1 RPA Workflow (in theory)

Planning includes for RPA implementation to the system using UiPath. The workflow is that

[1] UiPath retrieves all files from specified Google Drive folder

(https://drive.google.com/drive/folders/1Rw8THYvMOa_CTuS7ltOppOwUxjvE1hn4)



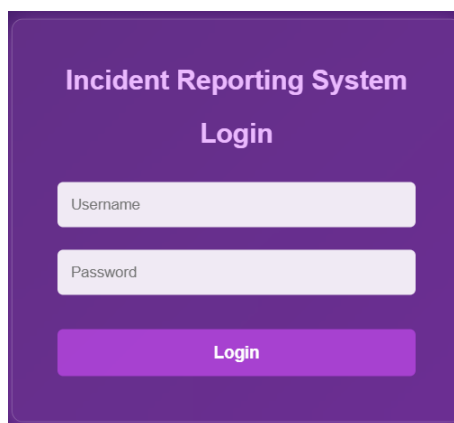
[2] For each file, UiPath downloads the file



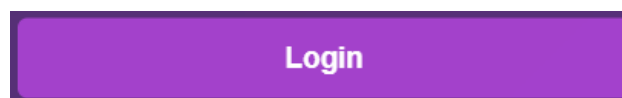
[3] UiPath fills in **username** and **password** in login.html of Incident Reporting System

Username: KaiFazli

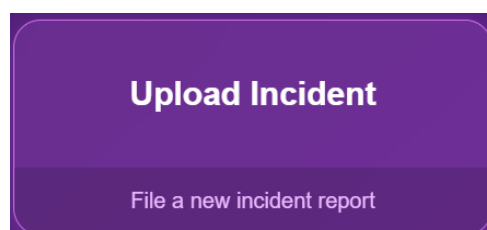
Password: 040324030801



[4] UiPath clicks “Login” button in login.html

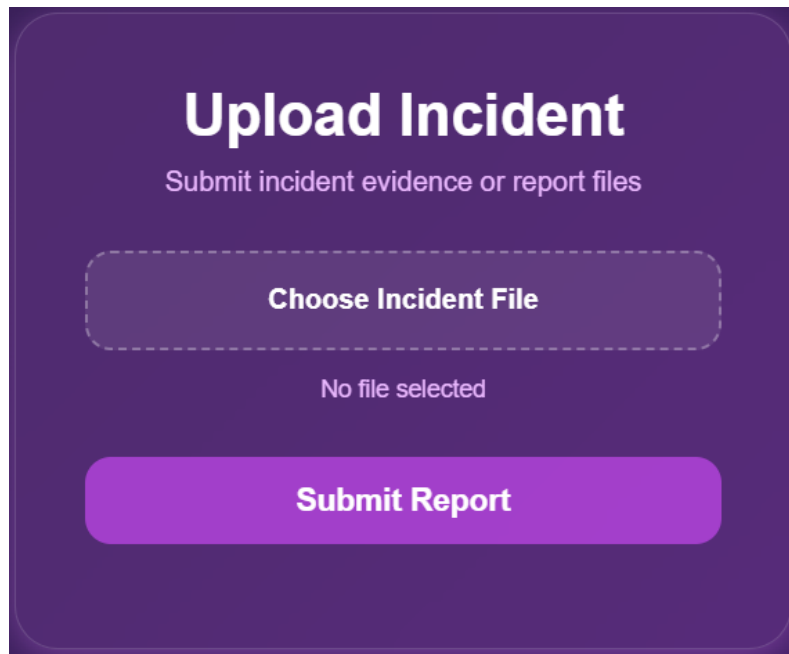


[5] UiPath clicks “Upload Incident” in main.html

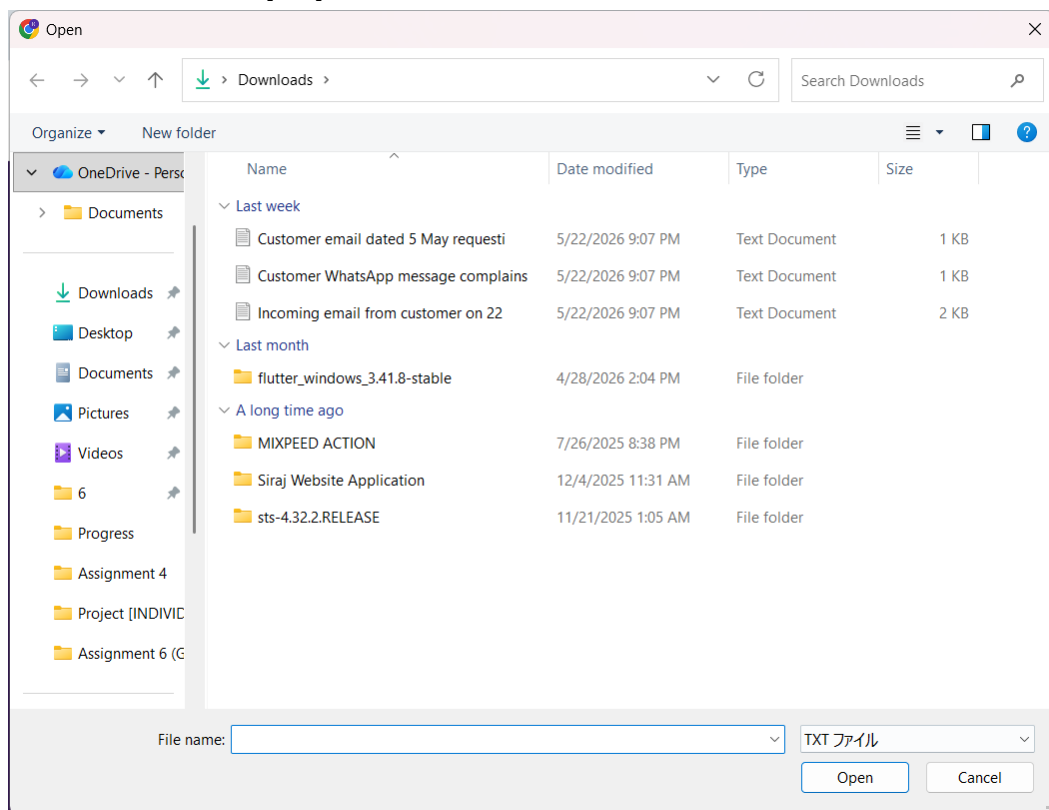


[6] For each file...

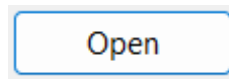
[6.1] UiPath clicks on “Choose Incident File” in upload.html



[6.2] UiPath clicks on downloaded .txt file



[6.3] UiPath clicks “Open” on file selection



[6.4] UiPath clicks on “Submit Report”

